Project Title

Smart Page Replacement Algorithm

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ABSTRACT

The Smart Page Replacement Algorithm is designed to improve memory management in operating systems by reducing page faults and utilizing available memory more efficiently. When the physical memory becomes full and a new page is requested, the operating system must select an existing page for replacement. Traditional algorithms such as First-In, First-Out (FIFO) and Least Recently Used (LRU) have certain limitations in handling different memory access patterns. The proposed Smart Page Replacement Algorithm aims to overcome these limitations by considering factors such as page access frequency, recency of use, and page usage history to identify the least useful page for replacement. The system simulates memory access requests and compares the performance of FIFO, LRU, and the proposed smart algorithm using metrics such as page faults, page hits, hit ratio, and execution time. A visualization interface can be developed to display memory frames and replacement decisions, making the working process easy to understand. The experimental results will help evaluate the effectiveness of the proposed approach under different workloads. This project aims to provide an efficient and user-friendly solution for understanding and improving page replacement strategies in operating systems. The proposed algorithm's performance will be validated through experimental comparison rather than assuming it always outperforms existing methods.

Signature of the Guide